

Cloud Computing: On-Prem vs Cloud, Scalability and Elasticity

Training / Learning Documentation

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1. On-Prem (On-Premises)

Definition:

Systems and servers are **owned, installed, and managed inside your organization's physical location.**

Key idea: You control everything.

Example:

Your company buys servers, keeps them in a server room, manages power, cooling, security, and maintenance.

In one line:

On-Prem = you build and run your own infrastructure.

2. Cloud

Definition:

Computing resources (servers, storage, etc.) are **provided over the internet by a third-party provider** like Microsoft Azure.

Key idea: You rent instead of owning.

Example:

Instead of buying servers, you create virtual machines online and use them as needed.

In one line:

Cloud = you use infrastructure provided by someone else.

3. Scalability

Definition:

The ability of a system to **handle increased workload by adding more resources**.

Important: Scaling can be:

- **Up (vertical):** Increase power of one machine (more RAM/CPU)
- **Out (horizontal):** Add more machines

Example:

Your app gets more users → you add more servers.

In one line:

Scalability = ability to grow when demand increases.

4. Elasticity

Definition:

The ability to **automatically increase AND decrease resources based on demand**.

Key difference from scalability:

- Scalability focuses on **growing**
- Elasticity focuses on **growing AND shrinking dynamically**

Example:

- Peak time → 2 servers → 5 servers
- After peak → back to 2 servers automatically

In one line:

Elasticity = automatic scaling up and down as needed.

Example: how the same workload behaves in a traditional setup vs cloud

TaxSmile -- IT Returns -- On Prem -- 2 Servers now -- Apr till Aug -- Peak Time

Arch -- Load Testing -- Stress Testing -- found 3 More Servers need

Get Approvals from Stake Holders -- Directors, CTO, COO

Place the order from Procurement Team -- Germany -- 3 weeks

Retrofit the server -- Network Engineers -- 24/7 -- 3 Teams + 1 Team (salary from the company)

Power Supply to the server room (company)

Rent for the Room- by the company

cannot return the 3 new servers back after the peak time

TaxSmile -- IT Returns -- Azure Cloud -- 2 VMs initial -- Apr till Aug -- Peak Time

in Text Box -- change from 2 to 5 -- Save

after peak time -- change from 5 to 2 -- save

Scaling – Elastic

Explanation:

1. On-Prem (your own servers)

Situation

- You currently have **2 servers**
- From **April to August (peak time)**, demand increases
- After testing, you realize: **you need 3 more servers**

What happens next (step-by-step reality)

1. Approvals

You must get permission from:

- Directors
- CTO
- COO

This takes time because it involves budget and decisions.

2. Procurement

- Order placed through procurement team
- Servers coming from Germany
- **Takes ~3 weeks**

👉 During this time, your system is still under pressure.

3. Setup (Retrofit)

- Network engineers install and configure servers
- Work happens **24/7 with multiple teams**
- Company pays salaries for all teams

4. Infrastructure Costs

You also need:

- Power supply (electricity, UPS, cooling)
- Physical server room
- Rent for space

👉 All paid by the company continuously

5. Big Problem (after peak time)

- Peak ends in August

- Now you don't need those extra 3 servers

✗ But:

- You **cannot return them**
- They sit idle
- Still consuming power, space, maintenance cost

Summary of On-Prem

- Slow setup
- High upfront cost
- Heavy operational effort
- Wastage after peak

2. Cloud Approach (using Microsoft Azure)

Situation

- Start with **2 virtual machines (VMs)**
- Same peak period: April to August

What happens here

During Peak

- You simply:
 - Change value from **2 → 5**
 - Click **Save**

👉 Cloud creates 3 more virtual machines instantly
No approvals, no shipping, no installation

After Peak

- Change from **5 → 2**
- Click **Save**

👉 Extra machines disappear

👉 You stop paying for them

Key Feature: Elastic Scaling

“Elastic” means:

- Resources **expand when needed**
- **Shrink when not needed**

Like a rubber band, it stretches and relaxes based on demand.

Final Comparison (core idea)

On-Prem

- Physical servers
- Time-consuming (weeks)
- Fixed capacity
- Extra resources wasted after use

Cloud

- Virtual servers (VMs)
- Instant scaling (minutes)
- Pay only for usage
- No waste after peak